Mobile Computing

Maps

Using GoogleMaps

- ThislecturefocusesonusingGoogle Maps inside an Android app
- AlternativesExist:
  - OpenStreetMaps
  - http://www.openstreetmap.org/
- Ifyousimplywanttodisplaya"standard Google map" from your app …

ClickerQuestion

- WhatAndroidcomponentwecanuse from our app to display a map of a location?

A. Activity

B. BroadcastReceiver

C. ContentResolver

D. Intent

E. Service

DisplayingStandardGoogleMap

- CreateURIbasedondesiredlocation.
- Fromlocationsample:

```
// show current location on map
public void showMap(View view) {
    if (lastKnownLocation != null) {
        // Create a Uri from an intent string.
        // // Use last known location.
        double lat = lastKnownLocation.getLatitude();
        double lng = lastKnownLocation.getLongitude();
        String locationURI = "geo:" + lat + "," + lng;
        Uri uriForMappingIntent = Uri.parse(locationURI);
```

CreateandFireIntent

```
// Create an Intent from gmmIntentUri.
// Set the action to ACTION_VIEW
Intent mapIntent = new Intent(Intent.ACTION_VIEW,
        uriForMappingIntent);

// Make the Intent explicit by setting the Google Maps package.
// If want to use user's preferred map app, don't do this!
mapIntent.setPackage("com.google.android.apps.maps");

// Attempt to start an activity that can handle the Intent
if (mapIntent.resolveActivity(getPackageManager()) != null) {
    startActivity(mapIntent);
}
```

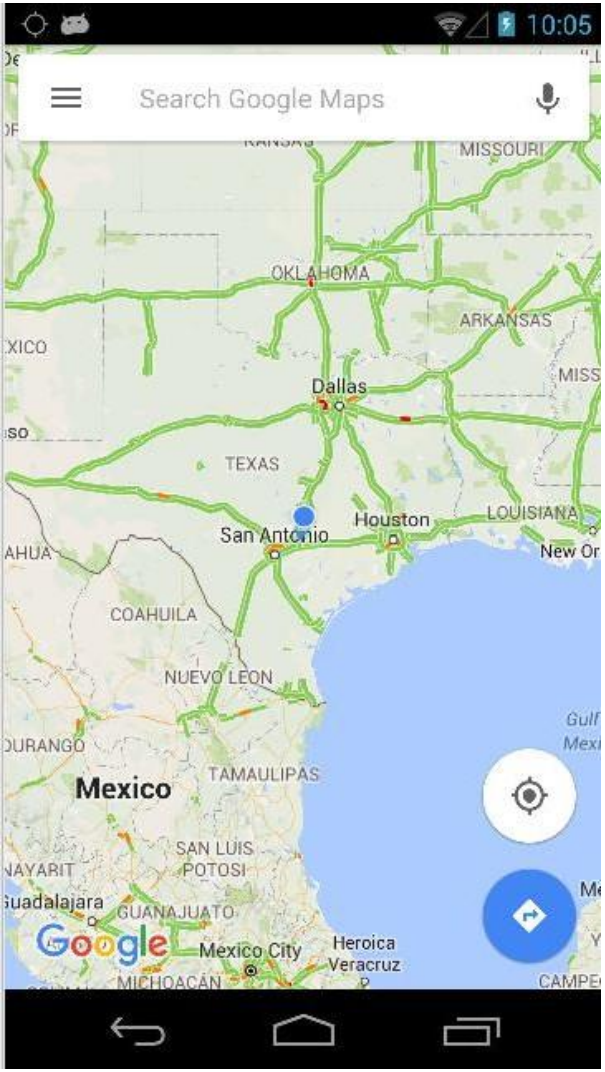
AdjustingZoomLevel

- SpecifyZoomLevelforGoogleMap
- ZoomLevelsinGoogleMaps,0-21
  - 0isthewholeearth
  - 21individualbuildings

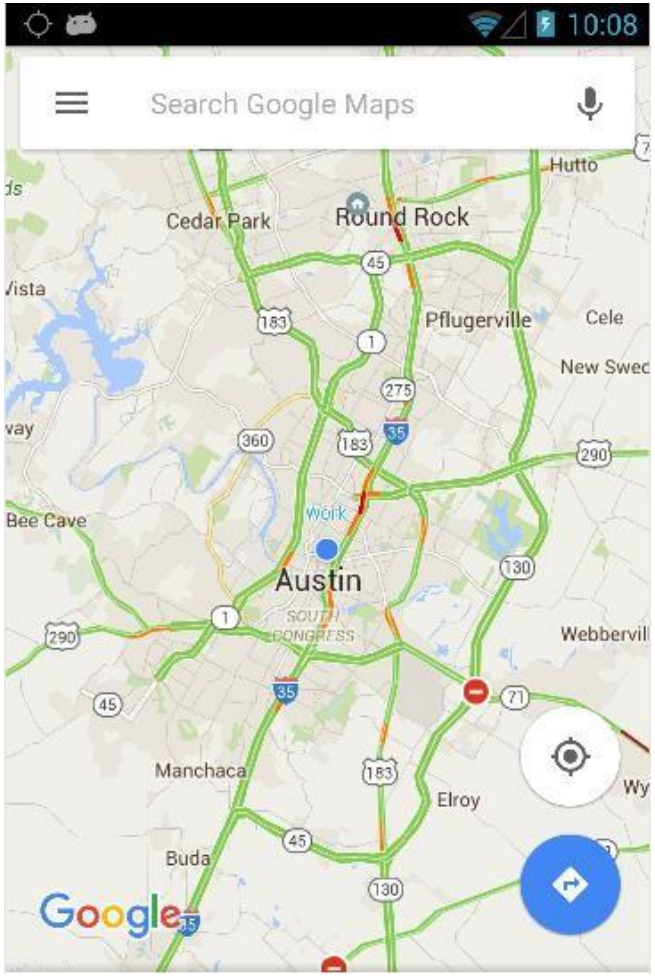
```
// show current location on map
public void showMap(View view) {
    if (lastKnownLocation != null) {
        // Create a Uri from an intent string.
        // // Use last known location.
        double lat = lastKnownLocation.getLatitude();
        double lng = lastKnownLocation.getLongitude();
        String locationURI = "geo:" + lat + "," + lng;
        locationURI += "?z=10";
        Uri uriForMappingIntent = Uri.parse(locationURI);
```

ZoomComparisons

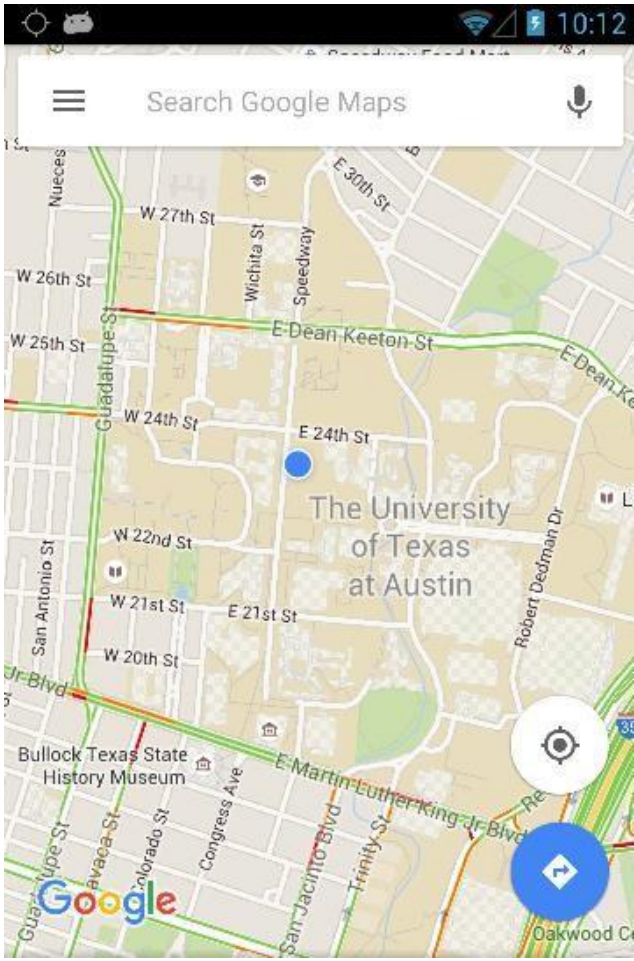
zoom=5



zoom=10



zoom=15

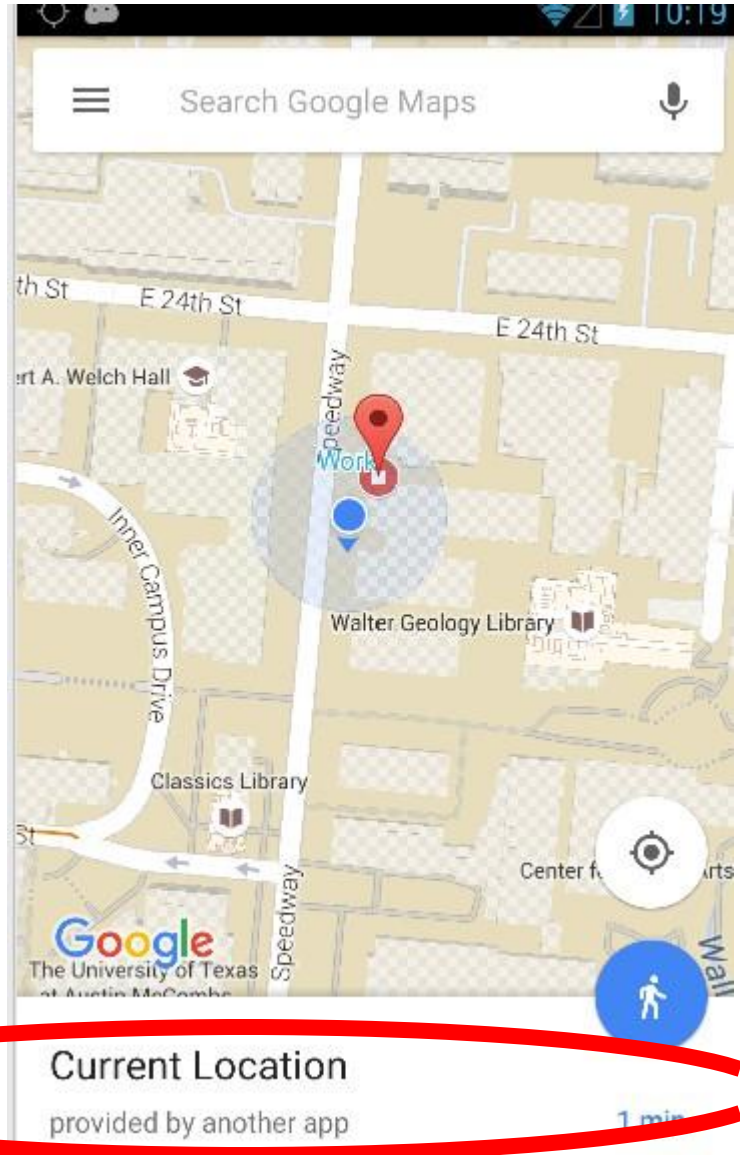


SearchingforLocationsandLabel

- CancreateURIsthatsearchforalocation and provide a label

```
locationURI = "geo:0,0?q="
        + lat + "," + lng
        + "(Current Location)";

uriForMappingIntent
        = Uri.parse(locationURI);
```



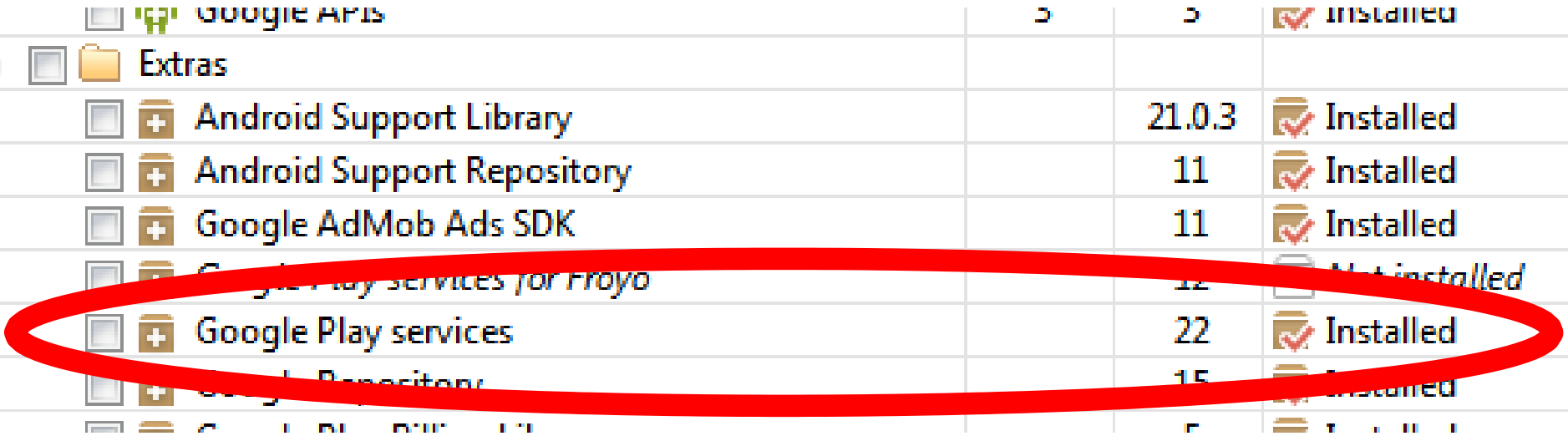
INCORPORATINGMAPSIN YOUR APPS

Using GoogleMaps

- NotstandardAndroid
- RequiresanAPIkeyfromGoogle
- https://developers.google.com/maps/documentation/android/
- requiredtouseMapViewclassor MapFragments
- MustaddareferencetotheMapsAPI Key in each MapView (xml or code)

UsingGoogleMapsAPIv2

- ForAndroid,GoogleMapsAPIv2partof the *Google Play Services sdk*
- DownloadviaSDK Manager



IncludeGooglePlayServicesinManifest

- TomakeuseofGooglePlayServicesadd data to manifest
- GooglePlayServiceshasahostofnon standard android tools
  - "simplelocationAPI"
  - "activity recognition"

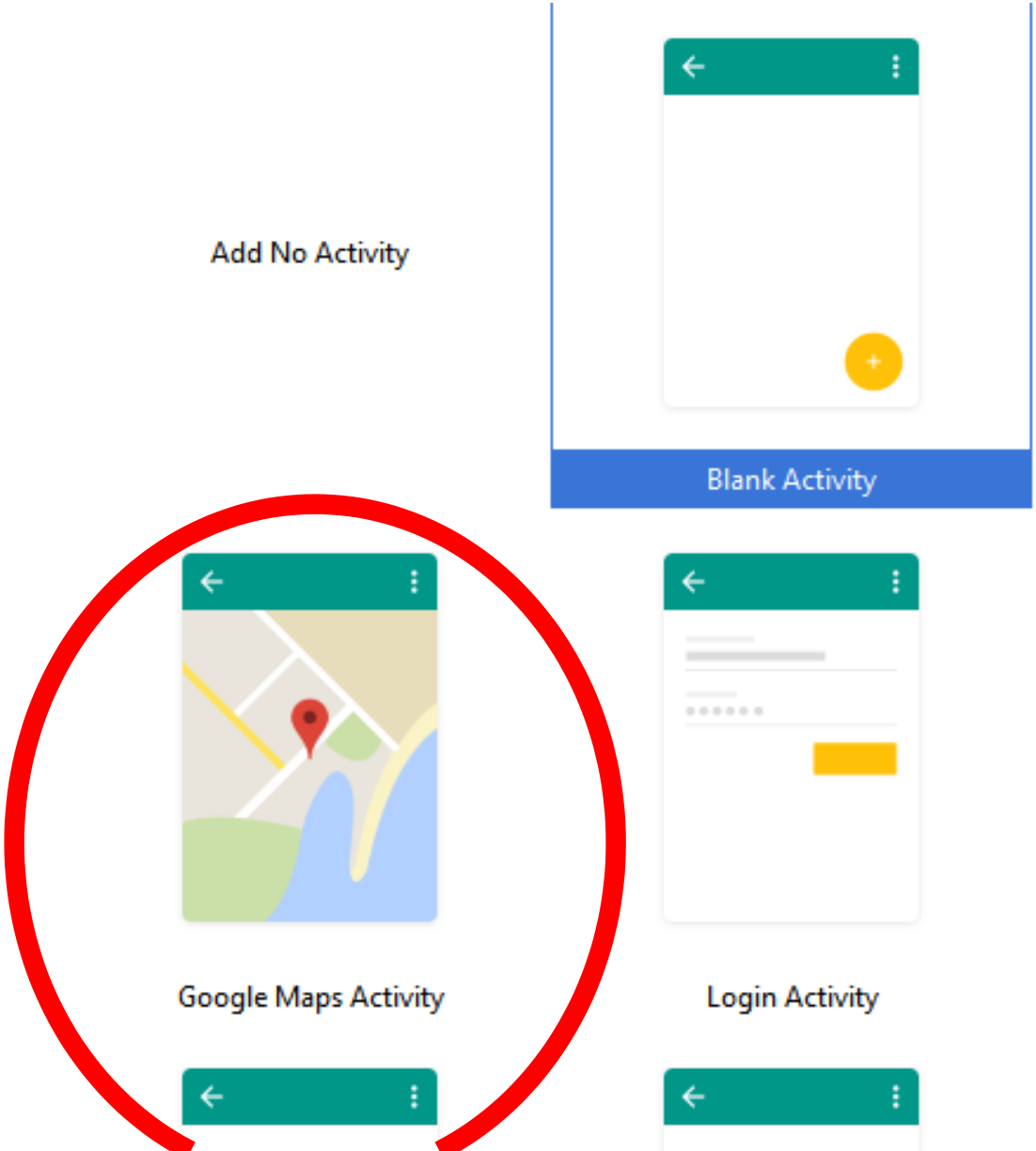
```
<meta-data
    android:name="com.google.android.gms.version"
    android:value="@integer/google_play_services_version" /
```

ObtaininganAPIKey

- MostwebAPIsrequireakeytouse
  - afewdonotsuchastheYahoofinance API
- SamewithGoogleMapsAPI
- New way of obtaining keys via Android StudioandGoogleDevelopersconsoleis mostly painless
- Old way of obtaining the key required someknowledgeregardinghowappsare published and was painful

NewWaytoGetMapsAPIKey

- CreateaGoogle MapsActivityin Android Studio



NewWaytoGetMapsAPIKey

- Lookatgoogle_maps_api.xmlfileinnew project

```
<!--
TODO: Before you run your application, you need a Google Maps API key.

To get one, follow this link, follow the directions and press "Create" a

https://console.developers.google.com/flows/enableapi?apiid=maps_android
```

GotoDeveloperConsole

Google Developers Console

Register your application for Google Maps Android API in Google Developers Console

Google Developers Console allows you to manage your application and monitor API usage.

Select a project where your application will be registered

You can use one project to manage all of your applications, or you can create a different project for each application.

Create a new project

Continue

CreateKey

API API Manager

Overview

Credentials

Credentials

Create Android API key

Name

Android key 1

Restrict usage to your Android apps (Optional)

Android devices send API requests directly to Google. Google verifies that each request comes from an Android app that matches a package name and SHA-1 signing-fingerprint name that you provide. Get the package name from your AndroidManifest.xml file. Use the following command to get the fingerprint. Learn more

```
keytool -list -v -keystore mystore.keystore
```

API key

Here is your API key

AIzaSyDWtuMQOMPTWWAajQz27BaU

OK

LastStep

- CopyAPIKeyintogoogle_maps_api.xml file

```
https://developers.google.com/maps/documentation/android/sta

Once you have your key (it starts with "AIza"), replace the
string in this file.
-->
<string name="google_maps_key" templateMergeStrategy
    ="preserve" translatable="false">YOUR_KEY_HERE</string>
</resources>
```

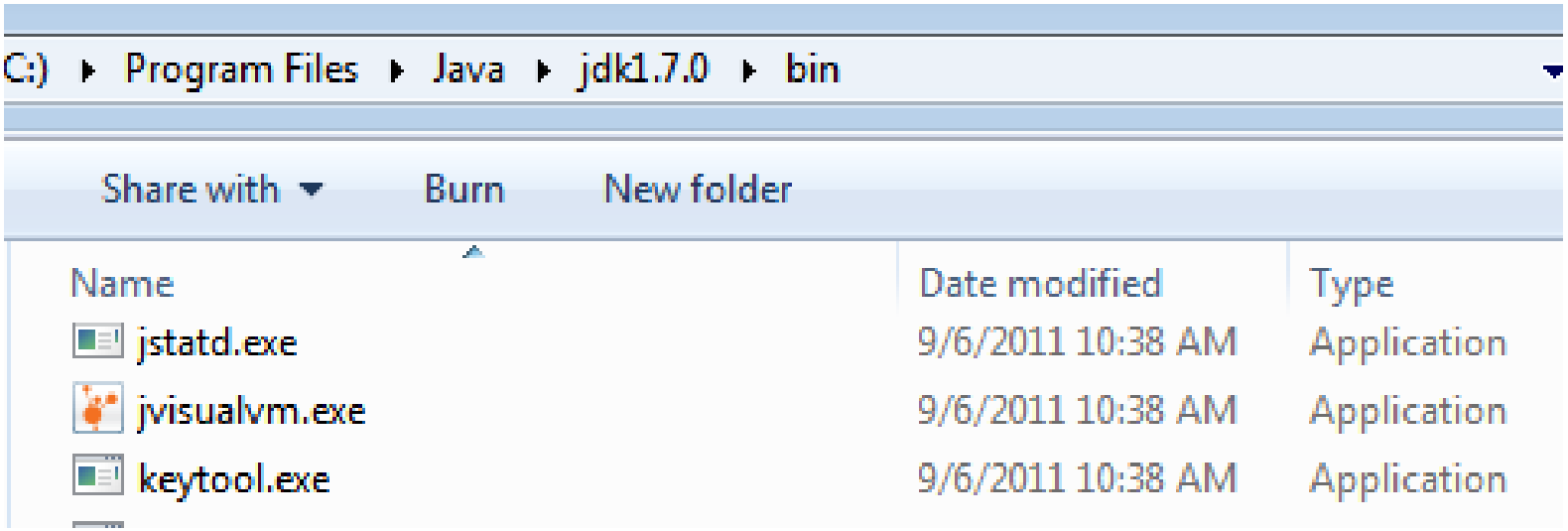
OLDPAINFULWAY

Signing Apps

- deployingappsontheGooglePlayrequires signing the app with a certificate
- developmentanddebuggingusesan automatic key creation process
  - invisibletous
- Inreleasemodeyoumustcreateyourown private key to sign apps
  - useofkeytoolprogramfromJavaSDK
- http://developer.android.com/guide/publishing/app-signing.html

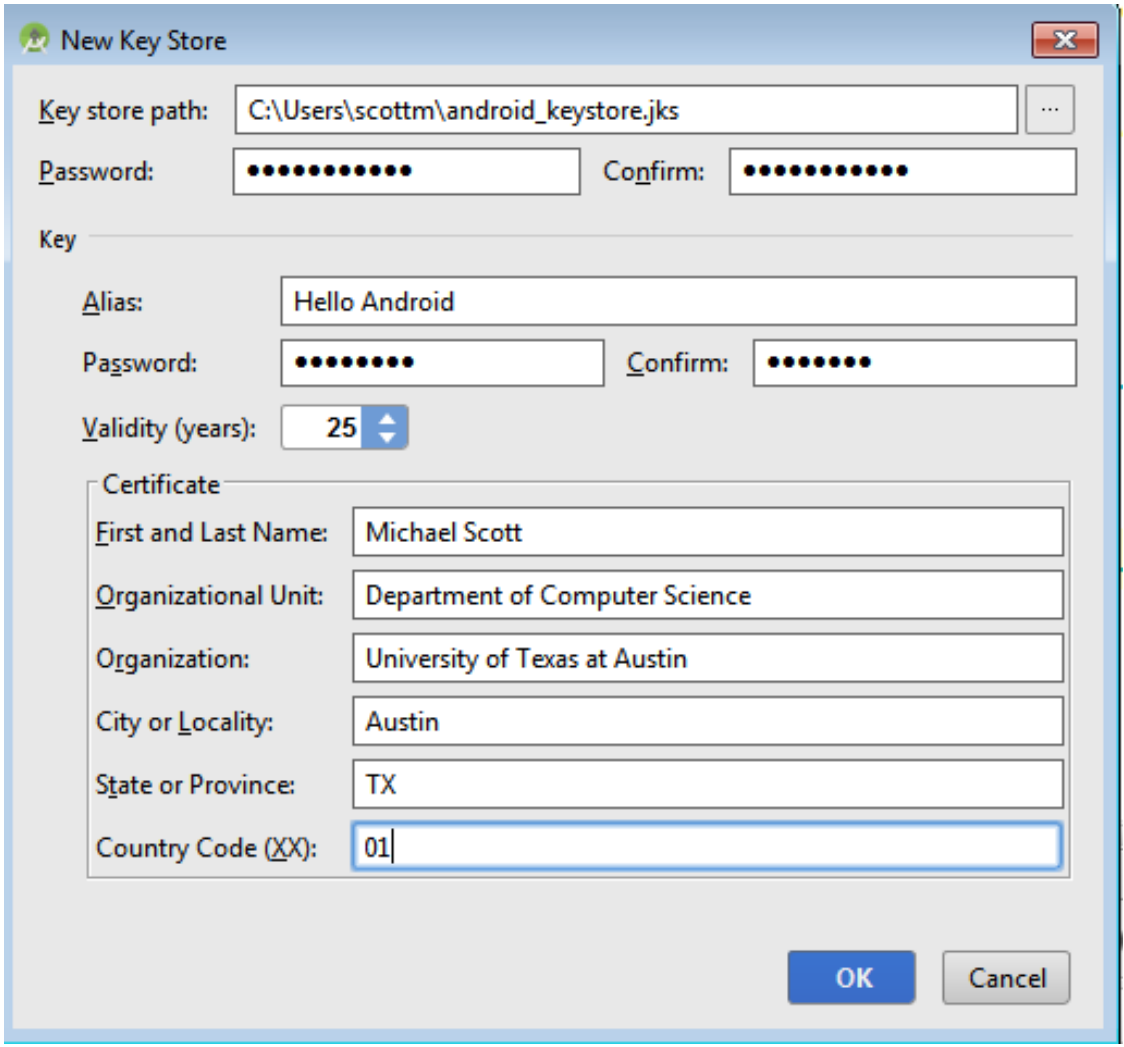
Signing Apps

- AJavaKeystoreisafile(orfiles)that stores security certificates
- IncludedintheJDK(JavaDevelopment Kit) is the keytool program
- Usedtocreatemanipulatethekeystore



SigningAppsviaAndroidStudio

- AndroidStudio provides a GUI to run keytool for you
- Build ->Generate SignedAPK



keystorefile

```
  0: FE ED FE ED 00 00 00 02  00 00 00 01 00 00 00 01  þíþí............
 10: 00 0D 68 65 6C 6C 6F 20  61 6E 64 72 6F 69 64 00  ..hello android.
 20: 00 01 4B CC 5B B7 F0 00  00 05 00 30 82 04 FC 30  ..KÌ[·ð....0▮.ü0
 30: 0E 06 0A 2B 06 01 04 01  2A 02 11 01 01 05 00 04  ...+....*.......
 40: 82 04 E8 85 46 05 B0 CC  67 08 BD A4 F6 40 25 B6  ▮.è▮F.°Ìg.½¤ö@%¶
 50: B3 1D A6 BF 94 03 52 7A  31 08 B2 68 50 5F 34 FB  ³.¦¿▮.Rz1.²hP_4û
 60: 9F 72 D7 0E 1B 75 3A 7A  72 A9 BE 22 D5 4F D5 A2  ▮r×..u:zr©¾"ÕOÕ¢
 70: A6 32 BC AB 44 39 00 4F  11 43 89 55 04 38 9B 24  ¦2¼«D9.O.C▮U.8▮$
 80: AA 89 11 B7 A6 EF E3 C9  E6 48 9B 10 27 B2 A4 04  ª▮.·¦ïãÉæH▮.'²¤.
 90: AC 97 73 3A D0 1D 7F 76  FC 22 3D A7 AA 9C 49 27  ¬▮s:Ð.▮vü"=§ª▮I'
 A0: CD B0 BE B0 CA 1F 83 5C  81 72 66 E5 A5 F9 1F B3  Í°¾°Ê.▮\rfå¥ù.³
 B0: 67 16 A5 9C F0 FA 43 49  27 3B DD E0 B4 6A 50 3E  g.¥▮ðúCI';Ýà´jP>
 C0: 80 91 10 FC E3 03 29 29  34 B8 9E A1 D9 34 41 C5  ▮´.üã.))4¸▮¡Ù4AÅ
 D0: CC A9 DD 66 B7 48 63 1E  AE 66 E2 35 32 DA 21 29  Ì©Ýf·Hc.®fâ52Ú!)
 E0: F2 D3 A6 48 F8 B0 CB F7  CE F9 B9 1B 81 5C A0 01  òÓ¦Hø°Ë÷Îù¹.\ .
 F0: 74 44 F7 D8 DB 58 A4 B5  CD 96 0C BA F8 47 7F 14  tD÷ØÛX¤µÍ▮.ºøG▮.
100: 86 AF AF 90 D5 9A 49 5D  24 20 9F C8 88 E5 38 A8  ▮¯¯Õ▮I]$ ▮È▮å8¨
110: 9D 0D 6B DD BE E7 9B E5  AD AB C0 AD F0 E2 33 79  .kÝ¾ç▮å-«À-ðâ3y
120: 61 4B 79 9D 29 31 25 28  41 AE 6A 51 72 0D 32 E7  aKy)1%(A®jQr.2ç
130: 8A 9C 1F B8 5B BF B4 66  5E C0 1B EB 9A C5 3B AE  ▮▮.¸[¿´f^À.ë▮Å;®
140: 1C DC C4 F9 8E B7 BF 7B  C4 DF 18 3D E9 FE 5F C3  .ÜÄù▮·¿{Äß.=éþ_Ã
150: 6C D4 43 D8 74 F6 95 EE  31 6B 15 53 C4 DD D0 80  lÔCØtö▮î1k.SÄÝÐ▮
160: D3 87 5E 1A 98 B4 04 19  7C 3E 02 0E 48 56 65 32  Ó▮^.▮´..|>..HVe2
170: 0F 54 56 40 CD C4 8E AC  98 97 43 A2 3A 7D DE A7  .TV@ÍÄ▮¬▮▮C¢:}Þ§
180: 7F F9 C5 3D 46 3C 13 39  82 D8 6E 03 15 DD CD CD  ▮ùÅ=F<.9▮Øn..ÝÍÍ
```

ObtaininganAPIKey

- ForGoogleMapsAPIv2
- Onekeytiedtoonesigningcertificate
- Samekeyusedforallinstancesofapp
- Normallysignappswithdifferent certificates
- If so differentAPI keys requiredif two differentappsusemapsandsignedwith different certificates

ObtaininganAPIKey

- To obtain a Google Maps API key we needthe*SHA-1*fingerprintofthesigning certificate
- Ashortformofthecertificatebasedon the SHA-1 hashing algorithm
- runkeytoolfromthecommandlineto pull out fingerprint of certificate

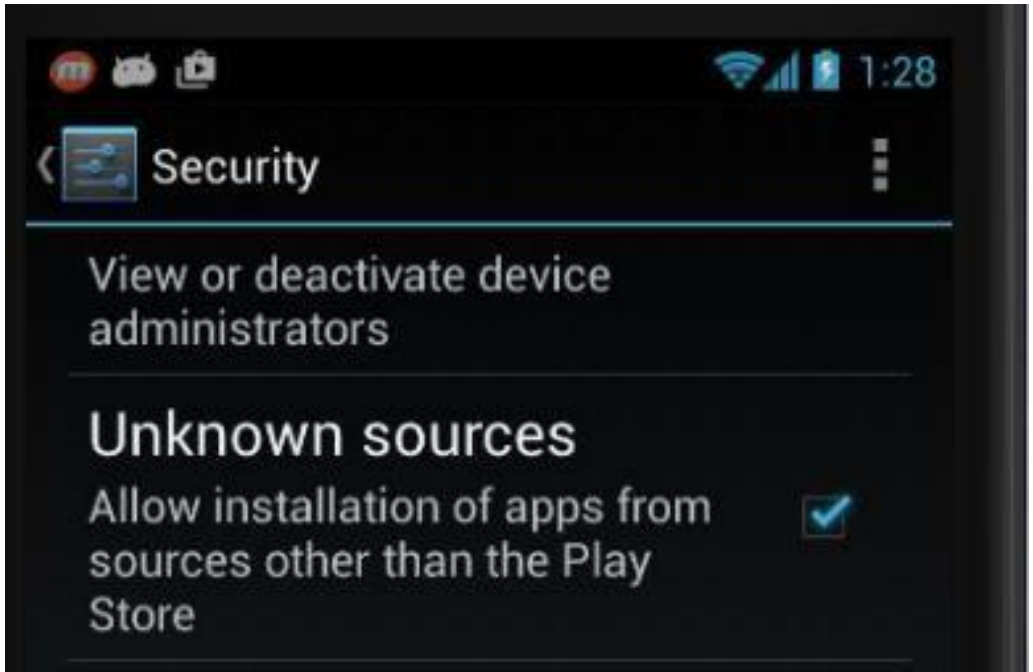
fingerprintviakeytool

- prompt>keytool-v-keystore <keystore_file_name>-alias <certificate_name>-storepass <keystore_password>-keypass <certificate_password>

```
C:\Users\scottm>keytool -list -v -keystore android_keystore.jks -alias s
Alias name: sample_certificate
Creation date: Feb 27, 2015
Entry type: PrivateKeyEntry
Certificate chain length: 1
Certificate[1]:
Owner: CN=Michael Scott, OU=Computer Science Department, O=University of
Issuer: CN=Michael Scott, OU=Computer Science Department, O=University o
Serial number: 6eee9bb2
Valid from: Fri Feb 27 12:57:48 CST 2015 until: Sat Feb 14 12:57:48 CST
Certificate fingerprints:
         MD5:  0F:E8:90:9E:32:23:AE:1A:CC:10:F2:71:B9:38:91:69
         SHA1: E5:07:2B:D9:CA:C7:4A:4B:5A:A4:B6:AC:60:49:A4:06:4E:31:F3:
         SHA256: 73:63:A0:25:94:DE:2B:4A:0B:AF:B9:5C:82:60:8E:F0:CA:2A:C
         Signature algorithm name: SHA256withRSA
         Version: 3
```

debugcertificate

- In development you are using a pre generateddebugkeystoretosignapps
- Happensbehindthescenes
- Securitysettingsondevice



debugcertificate

- possibletoobtainAPIkeytiedtoyour debug keystore
- worksindevelopment
- wouldneedtochangemanifestwith certificate used to sign the app

ObtainingtheAPI key

- TheSHA-1fingerprintisa20digit hexadecimal number
- UseGoogleAPIsconsoletoobtainkeyfor Maps
  - requiresGoogleaccount
- …andmustagreetothetermsof service.

UsingMapsAPIKeyinApp

- Mustaddkeytomanifest

```
<meta-data
    android:name="com.google.android.maps.v2.API_KEY"
    android:value="API_KEY"/>
```

- KEY POINT: Replace API_KEY in the secondlineonly(android:value)withthe API key you obtained in previous steps

DISPLAYINGAMAPINSIDEYOURAPP

GoogleMapsTermsofService

Google Maps/Google Earth APIs Terms of Service

☆☆☆☆☆

Thank you for your interest in the Google Maps/Google Earth APIs. The Google Maps/Google Earth APIs are a collection of services that allow you to include maps, geocoding, places,and other Content from Google in your web pages or applications.

Last Updated: January 23, 2017

- SomeHighlights
  - mayincludeadsinfuture
  - Googlemaylimitnumberof transactions
  - YouwillnotusetheServiceorContentfororinconnection with (a) real-time navigation or route guidance; or (b) automatic or autonomous vehicle control.
  - MustincludeGooglePlayServicesAttributioninyour apps"legalnotices"

Permissions

- Recommended Permissions for manifest whenusingGoogle Mapsinsideyourapp

```
<uses-permission android:name="android.permission.INTERNET"/>
<uses-permission android:name="android.permission.ACCESS_NETWORK_STATE"/>
<uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE"/>
<uses-permission android:name="android.permission.ACCESS_COARSE_LOCATION"/>
<uses-permission android:name="android.permission.ACCESS_FINE_LOCATION"/>
```

DisplaySimpleMapin App

- Hello Map
- LikeHelloWorld,butlayoutfilebecomes:

```
<?xml version="1.0" encoding="utf-8"?>
<fragment xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/map"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:name="com.google.android.gms.maps.MapFragment"/>
```

- MoreonFragmentslater,buttheyarea UI component between Activities and GUI Widgets

HelloMapActivity

- Lotsofnewclasses!!

```
import android.app.Activity;
import android.os.Bundle;

import com.google.android.gms.maps.CameraUpdate;
import com.google.android.gms.maps.CameraUpdateFactory;
import com.google.android.gms.maps.GoogleMap;
import com.google.android.gms.maps.MapFragment;
import com.google.android.gms.maps.model.BitmapDescriptorFactory;
import com.google.android.gms.maps.model.LatLng;
import com.google.android.gms.maps.model.Marker;
import com.google.android.gms.maps.model.MarkerOptions;
```

SpecifyingLocations

- LatitudeandLongitude

```
public class SimpleMapActivity extends Activity {

    static final LatLng AUSTIN = new LatLng(30.287, -97.737);
    static final LatLng ARLINGTON = new LatLng(32.751, -97.083);
    private GoogleMap map;
```

GoogleMapobject
Usedformostinteractionswithmap.

onCreateforSimpleMapApp

```
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_simple_map);
    MapFragment mapFrag =
            ((MapFragment) getFragmentManager()
            .findFragmentById(R.id.map));
```

MoreonFragmentslater…

```
mapFrag.getMapAsync(new OnMapReadyCallback() {
            @Override
            public void onMapReady(GoogleMap googleMap) {
                map = googleMap;
                setMarkers();
                moveCamera();
            }
        });
```

addingMarkersinonCreate

- firstMarkerusesdefault,pinandhasa title, "Austin"

```
if (map != null) {
    Marker austin
            = map.addMarker(new MarkerOptions().position(AUSTIN)
            .title("Austin"));

    Marker arlington = map.addMarker(new MarkerOptions()
            .position(ARLINGTON)
            .title("Arlington")
            .snippet("Play Ball!!")
            .icon(BitmapDescriptorFactory
                    .fromResource(R.drawable.ic_launcher)));
```

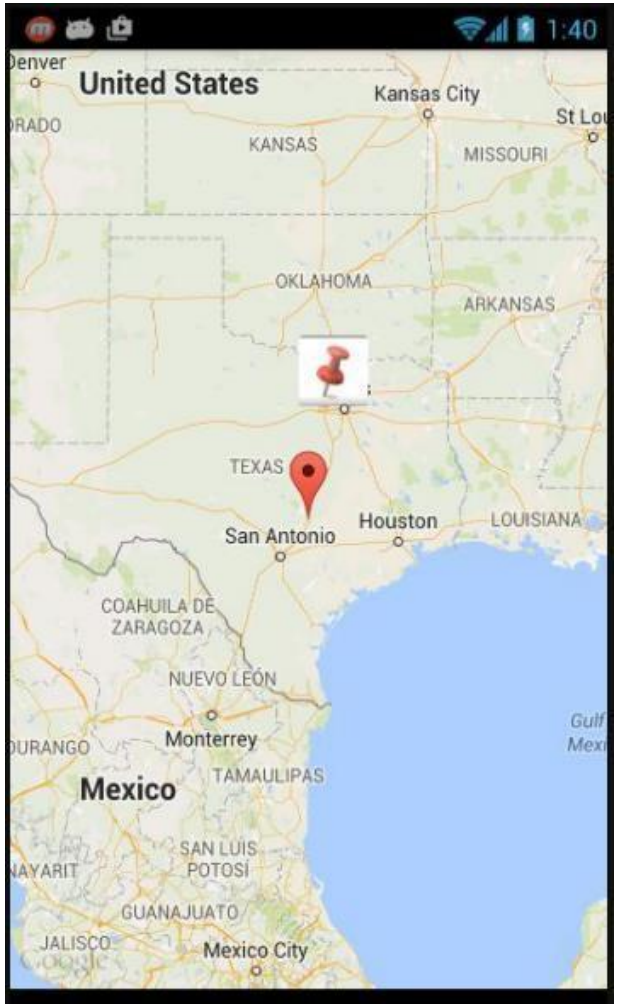
- secondMarkerusesadifferenticonand addstextafterthetitle

CenterandZoom

- Runningappasis produces this:
- Centeredwhere???
- Zoomedout
- Recall,zoomlevels0 to 21

```
CameraUpdate center=
        CameraUpdateFactory.newLatLng(AUSTIN);
CameraUpdate zoom=CameraUpdateFactory.zoomTo(5);

map.moveCamera(center);
map.animateCamera(zoom);
```



ResourcesforWorking withGoogleMaps

- https://developers.google.com/maps/documentation/android-api/intro

Google Maps APIs

Stack Overflow
Ask a question under the google-maps tag.

GitHub
Fork our samples and try them yourself.

Blog
Read the latest updates, customer stories, and tips.

Issue Tracker
Something wrong? Send us a bug report!

MapOptions

- AddingUI controls
- Handlingusergestures
- Handlinguser events
- moveandzoomcamera
- Drawonthe map
  - markers,informationwindows,shapes, overlays
- Heatmaps,MarkerClusters

Heatmaps

HIS AREA:
WEST CAMPUS
W 27th St
W 26th St
W 25th St
Graham Pl
W 24th St
W 23rd St
W 22 1/2 St
W 22nd St
W 21st St
W 20th St
W Martin Luther King Jr Blvd
E Martin Luther King Jr Blvd
E Dean Keeton St
E 24th St
E 21st St
E 30th St
E 32nd St
E 31st St
E 29th St
E 28th St
E 34th St
E 22nd St
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E 11th St
E 10th St
W 15th St
W 12th St
W 10th St
W 9th St
W 8th St
W 6th St
W 5th St
15th St W
Kingsbury St
N Lamar Blvd
Windsor Rd
Parkway
Baylor St
WEST END
DOWNTOWN
FINANCIAL DISTRICT
MEDICAL DISTRICT
UNIVERSITY OF TEXAS AT AUSTIN
CENTRAL EAST AUSTIN
EAST AU
CHERRYWOOD
Manor Rd
Duncan Ln
Keith Ln
Lafayette Ave
French Pl
Breeze Terrace
Alamo St
Coleto St
Chicon St
Poquito St
Chestnut Ave
Salina St
Leona St
Comal St
Waller St
Navasota St
San Bernard St
Angelina St
Cotton St
Catalpa St
Olive St
Juniper St
New York Ave
Pennsylvania Ave
S L Davis Ave
Rosewood
Sabine St
Red River St
San Jacinto Blvd
Congress Ave
Colorado St
Lavaca St
Guadalupe St
San Antonio St
Nueces St
West Ave
Rio Grande St
Pearl St
San Gabriel St
Leon St
Longview St
Brazos St
Trinity St
Robert Dedman Dr
Speedway
Wichita St
Robinson
Nueces
Sa
Ave
Blvd
N L
2
5
4
2
6
4
4
3
2
7
3
2
4
10
4
3
3
2
4
2
4
235B
236
235A
235B
35
234C
234B

MarkerClusters

- Too many markers canclutterthedisplay
- Especiallyonsmaller displays
- Markerclustersused to group together markers until user zooms in

TheMap Object

OLDSLIDES PREGOOGLEMAPSAPI2

DebugKey

- Portionofdebug.keystore

debug.keystore - Hex

```
   0: FE ED FE ED 00 00 00 02  00 00 00 01 00 00 00 01  þíþí............
  10: 00 0F 61 6E 64 72 6F 69  64 64 65 62 75 67 6B 65  ..androiddebugke
  20: 79 00 00 01 32 40 69 7D  D2 00 00 02 BC 30 82 02  y...2@i}Ò...¼0▌.
  30: B8 30 0E 06 0A 2B 06 01  04 01 2A 02 11 01 01 05  ¸0...+....*.....
  40: 00 04 82 02 A4 72 C5 15  FB DB 83 59 66 FD EC 56  ..▌.¤rÅ.ûÛ▌YfýìV
  50: 9B 6B 2A 92 92 12 AF 76  64 EF 64 D8 10 C2 FE 7C  ▌k*´´.¯vdïdØ.Âþ|
  60: CF 1D 1C 8D FF E7 9B 9B  F5 48 4C 9F E5 14 E5 9F  Ï..ÿç▌▌õHL▌å.å▌
  70: 11 DE 96 9D 5D 93 A8 9F  37 EC 4C 71 D3 A6 7A DD  .Þ▌]▌¨▌7ìLqÓ¦zÝ
  80: 00 90 DD E9 BF DF 57 28  CE B5 20 30 B6 0D 3B 48  .ÝéßW(Îµ 0¶.;H
  90: 77 95 67 0A A6 D0 32 C8  4E 99 89 23 AD FB 78 0C  w▌g.¦Ð2ÈN▌▌#-ûx.
  A0: DB CC A7 17 4D 1C AF A0  4B 20 E9 F9 21 60 59 05  ÛÌ§.M.¯ K éù!`Y.
  B0: 8C 9A 8B B4 3F 3D 8E EB  51 3D 6C 4F 26 2E 4B 5C  ▌▌▌´?=▌ëQ=lO&.K\
  C0: 34 74 71 7C 08 F9 54 02  CE F7 54 DF 3B 99 F9 4D  4tq|.ùT.Î÷TB;▌ùM
  D0: 9F 11 C7 A8 A1 E5 A6 BF  70 B9 70 53 98 FA 3C 5E  ▌.Ç¨¡å¦¿p¹pS▌ú<^
  E0: 80 7F C0 11 B0 41 B2 31  93 37 B2 1A 5F 96 65 28  €▌À.°A²1▌7².▌e(
  F0: 31 A5 3D E4 37 FE 33 49  A9 BF C7 FA A3 99 BA 17  1¥=ä7þ3I©¿Çú£▌º.
 100: 69 EB EC 70 D2 59 CE 7A  59 EC 0A 06 4E 83 2B 71  iëìpÒYÎzYì..N▌+q
 110: E2 5E 0E F0 0E 03 2A 35  CB 98 A3 51 52 25 50 EC  â^.ð..*5Ë▌£QR%Pì
 120: 4C A2 98 D0 D5 A3 DA 00  58 25 CA D9 FC F1 AF 3F  L¢▌ÐÕ£Ú.X%ÊÙüñ¯?
 130: 13 94 6B 59 14 42 5A E9  82 89 73 B3 49 C8 63 8D  .▌kY.BZé▌▌s³IÈc
 140: 6E 25 23 3E C9 A8 CC 36  E3 C6 35 9E 4F 07 4F EB  n%#>É¨Ì6ãÆ5▌O.Oë
 150: A0 D8 0E 1D 1A EC 88 0A  D4 69 21 2D B6 87 67 E9   Ø...ì▌.Ôi!-¶▌gé
 160: 5A 36 43 02 4E 57 AA 2E  49 9A 32 F4 5D 75 58 4A  Z6C.NWª.I▌2ô]uXJ
 170: A2 9F 9D 5B 56 70 A5 47  8D 55 31 74 49 DA F6 6B  ¢▌[Vp¥GU1tIÚök
 180: DC 3E E6 B1 00 F0 7F E3  43 30 E2 8E 72 4A DA 36  Ü>æ±.ð▌ãC0â▌rJÚ6
 190: 3B 6A BD 8C 0C F5 4C 50  1E 0C 52 36 CB 54 31 B7  ;j½▌.õLP..R6ËT1·
 1A0: 27 F9 10 90 D8 3A 9B FE  BC 70 CC 51 1B 2A 38 A3  'ù.Ø:▌þ¼pÌQ.*8£
 1B0: 5C CE FC 72 F3 88 32 4D  B5 48 C3 38 BB 64 FD 6E  \Îüró▌2MµHÃ8»dýn
 1C0: 0E 73 93 70 EB 9D 7B 35  5B B2 10 B1 C8 FE E4 BB  .s▌pë{5[².±Èþä»
```

GettingMD5Fingerprint

- usekeytoolprogram
- keytoolpartofJavaSDK
- keytool-list-aliasandroiddebugkey -keystore <path_to_debug_keystore>.keystore -storepassandroid-keypassandroid
- givesMD5fingerprintofthedebug certificate
- keytoolofJava1.7givesSHA1bydefault
  - use-vafterkeytool,before-list

DebugAPIKey

Thank you for signing up for an Android Maps API key!

Your key is:

```
0ceOzsn6afbLp8R8ZRl_9rizJTjrJIoa4w_VkCQ
```

This key is good for all apps signed with your certificate whose fingerprint is:

```
DF:6E:BD                                    0C:98:B3
```

Here is an example xml layout to get you started on your way to mapping glory:

```
<com.google.android.maps.MapView
    android:layout_width="fill_parent"
    android:layout_height="fill_parent"
    android:apiKey="0ceOzsn6afbLp8R8ZRl_9rizJTjrJIoa4w_VkCQ"
    />
```

HelloMapView

- BuildTarget-Google,notAndroid

☐ Android 2.3.3	Android Open Source Project	2.3.3	10
☑ Google APIs	Google Inc.	2.3.3	10
☐ EDK 1.1	Sony Ericsson Mobile Communicati	2.3.3	10

- MapViewnotastandardAndroidclass
  - partofGoogleMapsLibrary
  - addtomanifest

```
<uses-library android:name="com.google.android.maps" />
```

- mustalsoincludeINTERNETpermission andLOCATIONpermission

Aside-Permissions

- http://developer.android.com/reference/android/Manifest.permission.html

ACCESS_CHECKIN_PROPERTIES	Allows read/write access to the "properties" table in the checkin databa
ACCESS_COARSE_LOCATION	Allows an application to access coarse (e.g., Cell-ID, WiFi) location
ACCESS_FINE_LOCATION	Allows an application to access fine (e.g., GPS) location
ACCESS_LOCATION_EXTRA_COMMANDS	Allows an application to access extra location provider commands
ACCESS_MOCK_LOCATION	Allows an application to create mock location providers for testing
ACCESS_NETWORK_STATE	Allows applications to access information about networks
ACCESS_SURFACE_FLINGER	Allows an application to use SurfaceFlinger's low level features
ACCESS_WIFI_STATE	Allows applications to access information about Wi-Fi networks
ACCOUNT_MANAGER	Allows applications to call into AccountAuthenticators.
ADD_VOICEMAIL	Allows an application to add voicemails into the system.
AUTHENTICATE_ACCOUNTS	Allows an application to act as an AccountAuthenticator for the Account

MapView

- Atypeofviewforlayoutfile

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/mainlayout"
    android:orientation="vertical"
    android:layout_width="fill_parent"
    android:layout_height="fill_parent" >

    <com.google.android.maps.MapView
        android:id="@+id/mapview"
        android:layout_width="fill_parent"
        android:layout_height="fill_parent"
        android:clickable="true"
        android:apiKey="Your Maps API Key"
    />

</RelativeLayout>
```

MapActivity

- Createclassthatextendsmapactivity instead of Activity
- importcom.google.android.maps.MapActivity;
- mustimplementisRouteDisplayed method

```
@Override
protected boolean isRouteDisplayed() {
    return false;
}
```

- mustreturntrueifanykindofroute(to be followed) is displayed, per terms of use

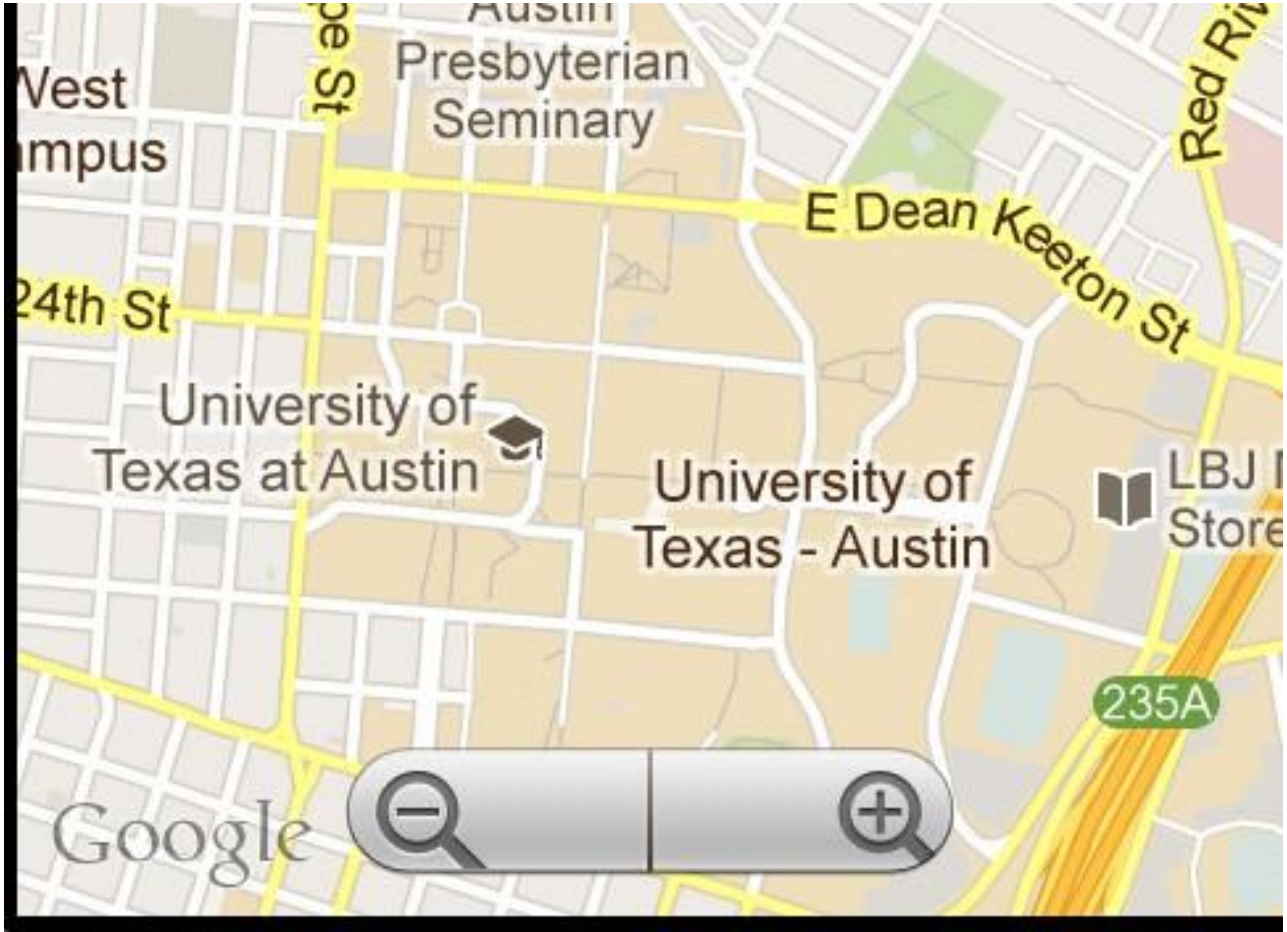
InstanceVarsandonCreate

- Addinstancevariablesandinitializein onCreate method

```
private LinearLayout linearLayout;
private MapView mapView;

/** Called when the activity is first created. */
@Override
public void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);

    setContentView(R.layout.main);
    mapView = (MapView) findViewById(R.id.mapview);
    mapView.setBuiltInZoomControls(true);

}
```

HelloMapView

- Runapp
- Displaysmapandallowspanningandzooming



CustomizingMap

- Easytodisplaymapandallowinteraction
- Customizewithmarkersandoverlays
- Overlays
  - usedtodisplayinformationontopofmap
  - simplechoice:ItemizedOverlayclass

ItemizedOverlay

```
public class HelloItemizedOverlay extends ItemizedOverlay {

    private ArrayList<OverlayItem> mOverlays;

    public HelloItemizedOverlay(Drawable defaultMarker) {
        super(boundCenterBottom(defaultMarker));
        mOverlays = new ArrayList<OverlayItem>();
    }


    public void addOverlay(OverlayItem overlay) {
        mOverlays.add(overlay);
     // inherited method to prepare overlays to be drawn
        populate();
    }
```

ItemizedOverlay

- populatemethodwillcallcreateItem
- definecreateItemandreturnvaluefrom the ArrayList instance var
- definesizemethodthatreturnsnumber of overlay items

```
@Override
protected OverlayItem createItem(int i) {
    return mOverlays.get(i);
}

@Override
public int size() {
    return mOverlays.size();
}
```

Adding Overlays

- InMapActivitycreateOverlayItem
- addtoHelloItemizedOverlay
- addtoMapView
- Needadrawableforthemarker
  - res/drawable
  - issuesdisplaygif format images onsomedevices

ChangestoHelloMapView

```
private LinearLayout linearLayout;
private MapView mapView;
private List<Overlay> mapOverlays;
private Drawable drawable;
private HelloItemizedOverlay itemizedOverlay;

/** Called when the activity is first created. */
@Override
public void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);

    setContentView(R.layout.main);
    mapView = (MapView) findViewById(R.id.mapview);
    mapView.setBuiltInZoomControls(true);

    mapOverlays = mapView.getOverlays();
    drawable = this.getResources().getDrawable(R.drawable.longhorn);
    itemizedOverlay = new HelloItemizedOverlay(drawable);
}
```

AddOverlayItems

- CreateGeoPointandusethesetocreate OverlayItems
- GeoPointbasedonmicrodegrees
  - latandlongtimes 1,000,000
- BuildOverlayItemsoutofGeoPointsand include strings for title and snippet text to display when drawable clicked

addOverlaysmethodinHelloMapView

```
private void addOverlays() {
    int[] parlin = {(int) (30.284882 * 1e6), (int) (-97.740127 *1e6)};
    int[] mcdonald = {(int) (30.671581 * 1e6), (int) (-104.022431 *1e6)};
    int[] marine = {(int) (27.835926 * 1e6), (int) (-97.050372 *1e6)};
    int[][] points = {parlin, mcdonald, marine};

    String[][] titlesAndSnippets = {{"UT",
        "Parlin Hall at The University of Texas at Austin"},
        {"McDonald",
            "McDonald Observatory - University of Texas - West Texas"},
        {"Marine Biology",
                "University of Texas Marine Sciennce Institure - Port Aransas"}}

    for(int i = 0; i < points.length; i++) {
        GeoPoint g = new GeoPoint(points[i][0], points[i][1]);
        OverlayItem oi = new OverlayItem(g,
                titlesAndSnippets[i][0], titlesAndSnippets[i][1]);
        itemizedOverlay.addOverlay(oi);
    }
    mapOverlays.add(itemizedOverlay);
}
```

Result

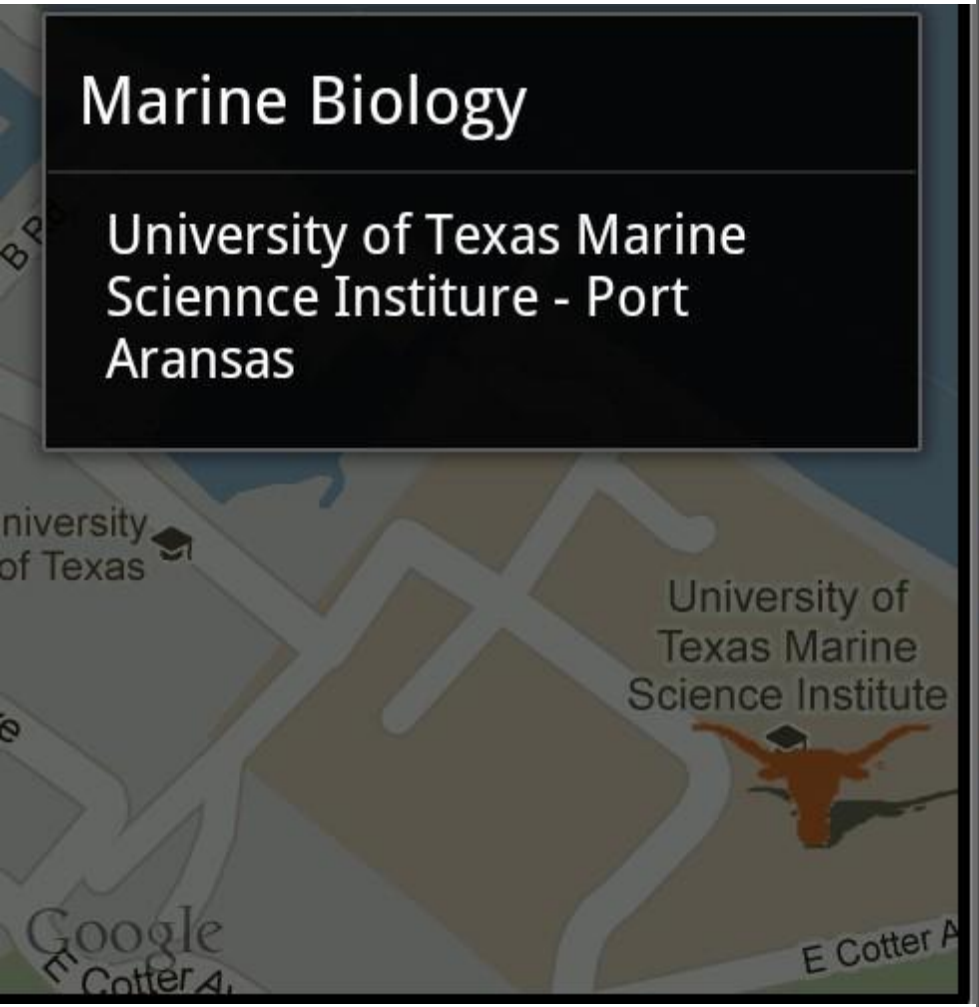
- one overlay withmultiple items
- based on locationswe added

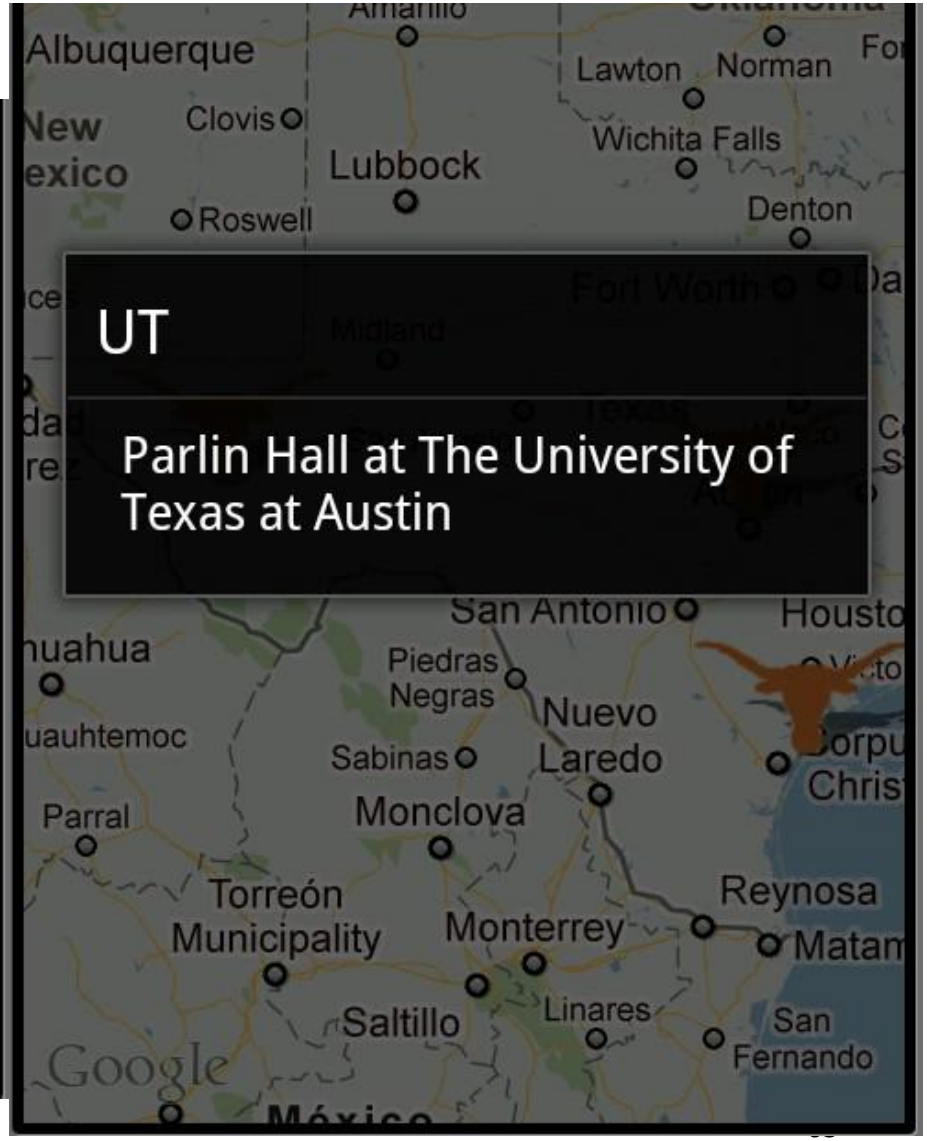
DisplayInformation

- Todisplayinformation(titleandsnippet) ofoverlayoverridetheonTapmethodin the ItemizedOverlay class

```
@Override
protected boolean onTap(int index) {
    OverlayItem item = mOverlays.get(index);
    AlertDialog.Builder dialog = new AlertDialog.Builder(mContext);
    dialog.setTitle(item.getTitle());
    dialog.setMessage(item.getSnippet());
    dialog.show();
    return true;
}
```

ResultsofClickingLonghorn





ReverseGeocoding

- Findaddressesfromlongitude/latitude
- Geocoder uses a backend that is NOT includedinthecoreandroidframework
- useisPresentmethodtocheckforservice

```
location=locationManager.getLastKnownLocation(
                        LocationManager.GPS_PROVIDER);doubl
e lat = location.getLatitude();
doublelng= location.getLongitude();

Geocodergc=newGeocoder(this,Locale.getDefault()); List<Address>
addresses = null;
try {
  addresses=gc.getFromLocation(lat,lng,5);//maxResults
}catch(IOExceptione){}
```

ForwardGeocoding

- Findlongitude/latitude(andmore)from addressorairportcode

```
Geocodergc=newGeocoder(this,Locale.US); List<Address>
addresses = null;
try{
  addresses=gc.getFromLocationName(
                    “713N.Duchese,St.,Missouri”,5);
}catch(IOExceptione){}
double lat = addresses.get(0).getLatitude();
doubleIng=addresses.get(0).getLongitude();
String zip = addresses.get(0).getPostalCode();
```

```
HelloMapView      Geocode is present: true
HelloMapView      lat: 38.7991079, long: -90.494416, zip: 63301
```

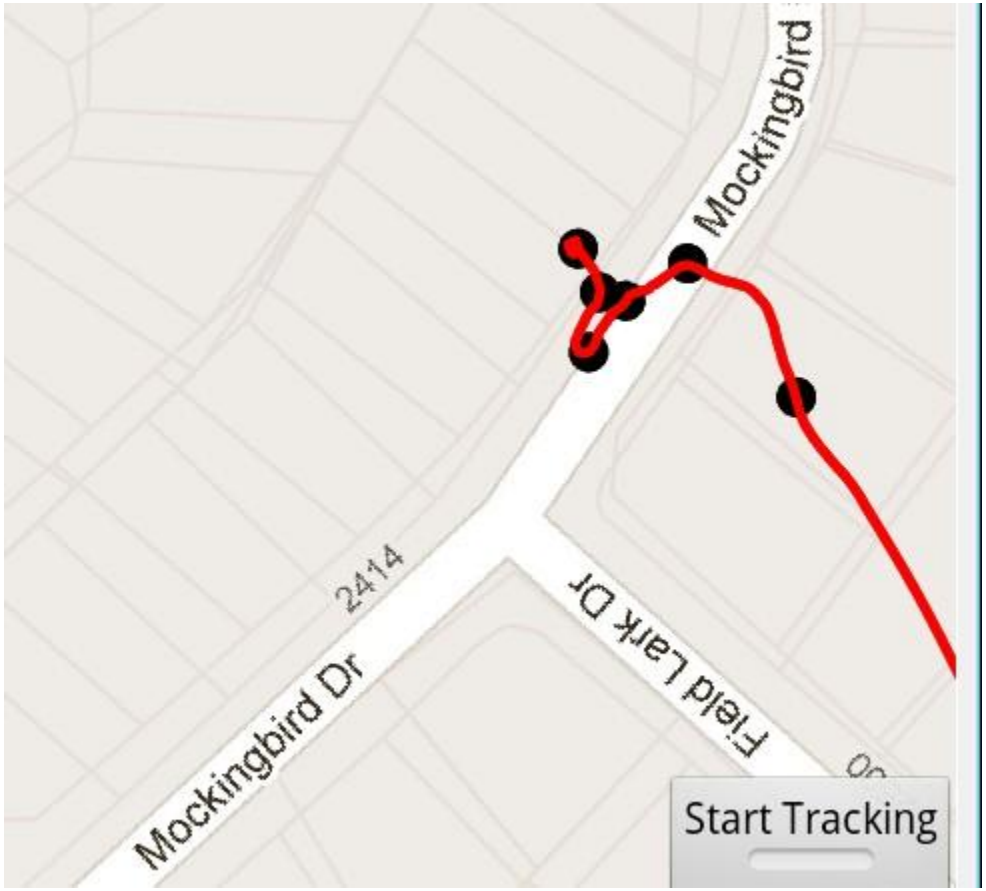
RecentChanges

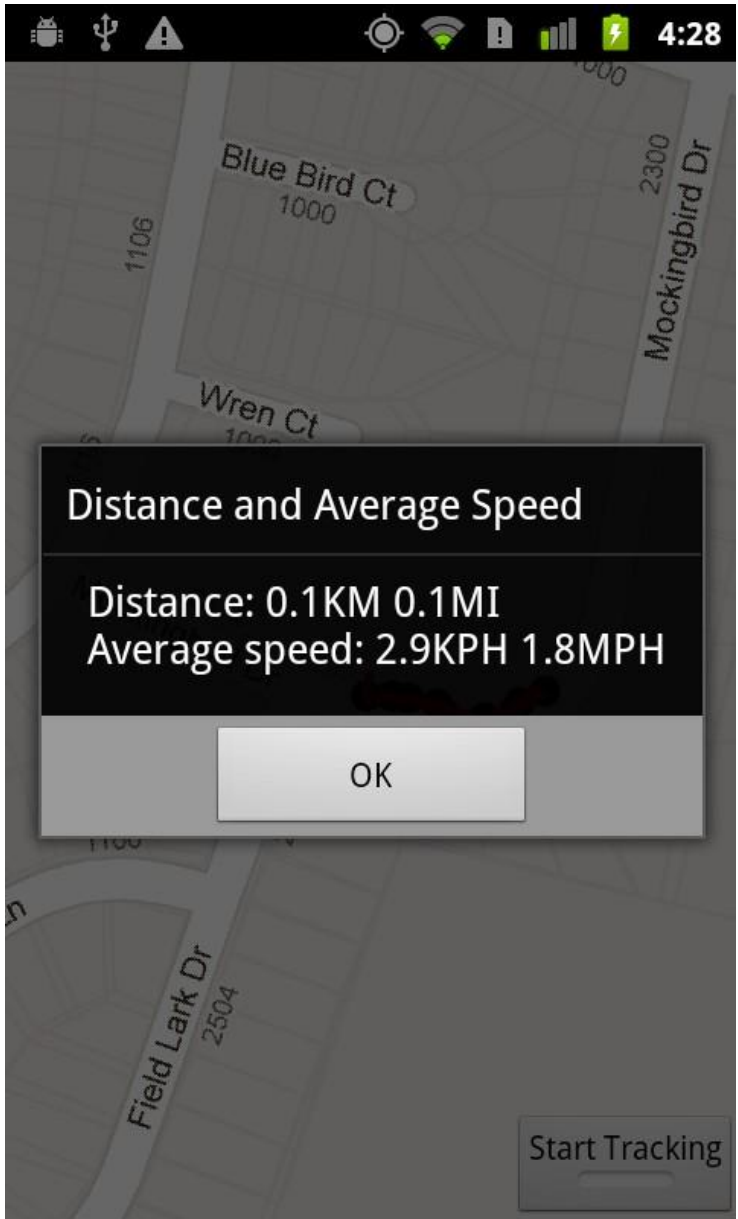
- GoogleMapsAPIversion2
  - somewhatnew,ReleasedDecember2012 as part of Google Play Services SDK
- features:
  - indoor maps
  - simplifiedlocationservices

MapsExample

- Route Tracker using Locations,MapActivity, MapView, and Google Maps
  - fromDeitelAFP-AADA
- SimilartoMapMyRide
  - popularappamong cyclistsandrunners

RouteTrackerApp





RouteTrackerApp

- using FrameLayout to stack components withthemostrecentlyaddedcomponent on top
- ToggleButtonatbottomtostartandstop route tracking
- MapViewaddedtoFrameLayout
- routeisanoverlaytomapwithpoints and lines connecting points

RouteTrackerClasses

RouteTracker

StartingActivity
dealswithLocationProvider

BearingFrameLayout

Displays MapView
rotatesbasedonbearing
from location

RouteOverlay

Overlaywithlocationpoints
(every 10th)
andlinesconnecting.
Convertslocationsto
GeoPoints.
Overloadsdraw

CriteriaClass

- SetcriteriaforselectingaLocationProvider

```
Criteria criteria = new Criteria();
criteria.setAccuracy(Criteria.ACCURACY_FINE);
criteria.setBearingRequired(true);
criteria.setCostAllowed(true);
criteria.setPowerRequirement(Criteria.POWER_LOW);
criteria.setAltitudeRequired(false);

locationManager =
        (LocationManager) getSystemService(LOCATION_SERVICE);

// get the best provider based on our Criteria
String provider = locationManager.getBestProvider(criteria, true);
```

GpsStatus.Listener

- RespondstochangesinGPSstatus
- ArewereceivingGPSfixes?
- Appdoesnottrackunlessthisis true

```
// determine whether we have GPS fix
GpsStatus.Listener gpsStatusListener = new GpsStatus.Listener() {
    public void onGpsStatusChanged(int event) {
        if (event == GpsStatus.GPS_EVENT_FIRST_FIX) {
            gpsFix = true;
            Toast results = Toast.makeText(RouteTracker.this,
                    getResources().getString(R.string.toast_signal_acquired),
                    Toast.LENGTH_SHORT);

            results.setGravity(Gravity.CENTER,
                    results.getXOffset() / 2, results.getYOffset() / 2);
            results.show();
        }
    }
};
```

SimulatingGPSData

- tosimulate changes in location in emulator
- GPSdataina file
  - GPSExchange Format (GPX)

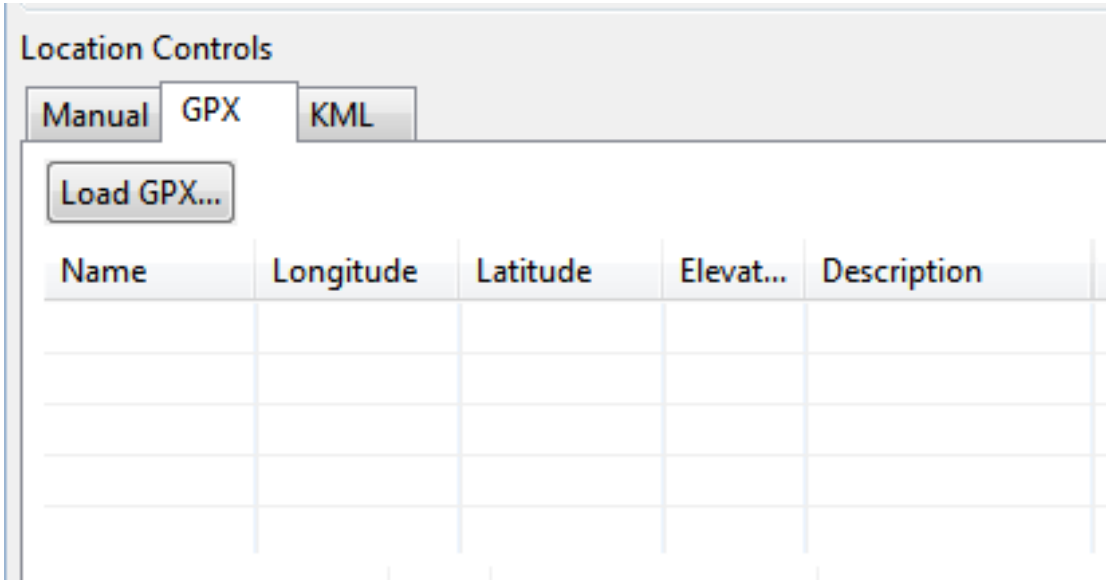
```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<gpx xmlns="http://www.topografix.com/GPX/1/1" xmlns:gpxtrx="http://
    <trk>
        <name>Position 1 to Position 517</name>
        <trkseg>
            <trkpt lat="42.37123092799624" lon="-71.43667976096056">
                <ele>1.0</ele>
                <time>2011-07-02T20:01:38.000Z</time>
                <name>Position 1</name>
                <sat>7</sat>
            </trkpt>
            <trkpt lat="42.37123092799624" lon="-71.43667976096056">
                <ele>1.0</ele>
                <time>2011-07-02T20:01:39.000Z</time>
                <name>Position 2</name>
                <sat>7</sat>
                <extensions>
                    <nmea:speed>0.0</nmea:speed>
                </extensions>
            </trkpt>
            <trkpt lat="42.37123092799624" lon="-71.43667976096056">
                <ele>1.0</ele>
                <time>2011-07-02T20:01:40.000Z</time>
                <name>Position 3</name>
                <sat>7</sat>
                <extensions>
                    <nmea:speed>0.0</nmea:speed>
                </extensions>
            </trkpt>
```

CreatingGPXFiles

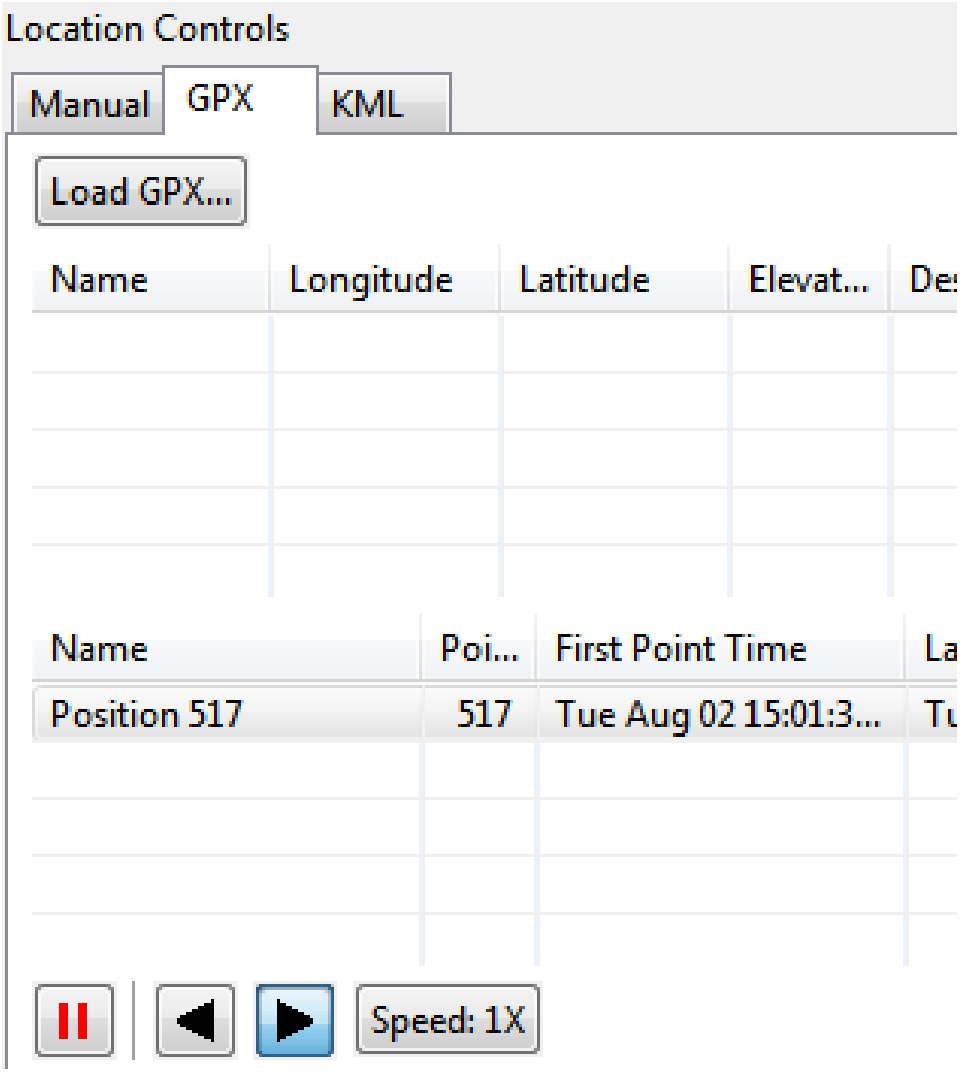
- Manyappsandprograms
- OneoptionforAndroiddevices
- GPSLogger
- gpsbabelto convert between variousGPSformats
  - –gpxhasdifferentversions

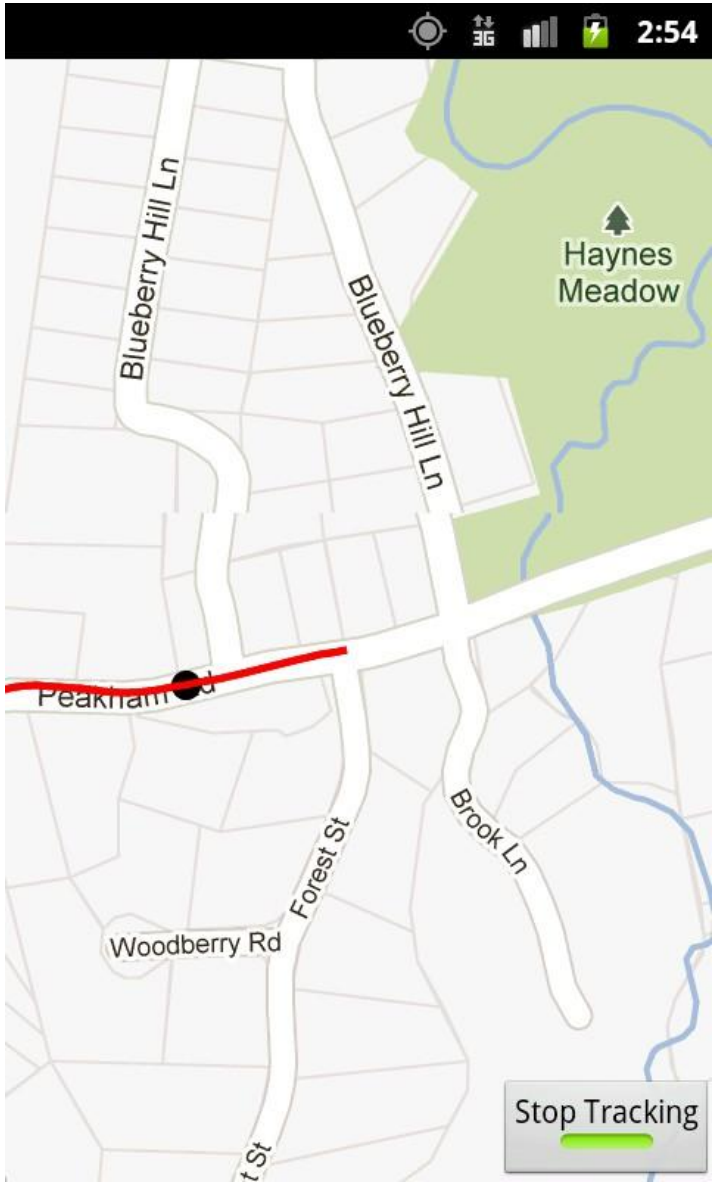
RunningGPXfilesin App

- DDMS
- EmulatorControlTab
- GPXTab
- Load



Running GPX


Location Controls
Manual
GPX
KML
Load GPX...
Name
Longitude
Latitude
Elevat...
Name
Poi...
First Point Time
Position 517
517
Tue Aug 02 15:01:3...
Speed: 1X


2:54
Blueberry Hill Ln
Blueberry Hill Ln
Haynes Meadow
Peakham
Forest St
Brook Ln
Woodberry Rd
Stop Tracking

Geocoding

- Findingaddressesfromlat/longandvice versa
- Reversegeocoding:findaddressfromlat and long
- Forwardgeocoding:findlatandlong from address